firstly, through an electric field which results from the charge carried by the protons, and secondly, through a nuclear field, which in some as yet undisclosed way ensures the internal cohesion of the nucleus. The electric field is responsible for the emission of gamma-ray photons, and the nuclear field is responsible for the emission of electrons, positrons and neutrinos.

Whether the connection between the nuclear field and the particles correlated with it here is as simple as that between the electric field and the photons, is, of course, open to doubt. As we shall see in due course, this relationship is probably a more complex one. But generally speaking, it is permissible to apply the analogy outlined above.

Thus we have obtained a fairly clear picture of the nucleus. We may interpret it as meaning that a sufficiently powerful super-microscope would show nuclei to be such as we have described them to be—built of two kinds of component parts, protons and neutrons. Therefore, every nucleus can be identified extremely simply, by two numbers: the number of its protons, and the number of its neutrons.

The mass of a nucleus is (to be sure, not quite exactly) equal to the combined number of its protons and neutrons; both particles have a mass of approximately 1 mass unit. On the other hand, the nuclear charge is possessed by the protons alone, each of which carries one elementary electric charge.

From this simple picture of the nucleus it follows, in general, automatically that different nuclear mass numbers can be associated with a given nuclear charge (atomic number). In other words, there are different varieties of atomic nuclei for the same chemical element; these different varieties are called the *isotopes* of the element.

The mass number of the nucleus indicates the sum total of the number of its protons and neutrons, whereas the atomic number shows the number of protons only. Hence, the number of the neutrons is simply the difference between the mass number and the atomic number. These two numbers, the essential characteristics of the atom, have already been introduced as a superscript and a subscript t6 the chemical symbol of the element. For instance, the notation ${}_{7}N^{14}$ means that this nitrogen nucleus consists of 7 protons and $14 - 7 = 7$ neutrons.

Let us now examine some of the simpler nuclei more closely. Disregarding the neutron, which is not usually grouped with the elements in the proper sense of the word, the simplest nucleus of an element is the proton itself. The proton is a hydrogen nucleus. Its symbol is ${}_{1}H^{1}$, expressing that it consists of 1 proton and $1 - 1 = 0$ neutron. The chart below shows schematic representations of the simplest nuclei; black dots stand for protons, circles for neutrons. Thus, the hydrogen nucleus ${}_{1}H^{1}$ is represented simply by a black dot. But as Urey discovered in 1932, there exists another nucleus, a heavy